ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | SUMMER SEMESTER, 2023-2024

DURATION: 3 HOURS | FULL MARKS: 150

Math 4441: Probability and Statistics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Let X and Y be random variables that take on values from the set $\{-1, 0, 1\}$. Find a joint probability mass assignment (i.e., joint probabilities assignment, $P[X = x, Y = y]$) for which X and Y are independent, and then, show that X and Y are really independent for the assigned joint PMF. 10 (CO1) (PO1)

   b) A point is chosen uniformly at random from the triangle that is formed by joining the three points $(0, 0)$, $(0, 1)$ and $(1, 0)$ (units measured in centimetre). Let X and Y be the co-ordinates of a randomly chosen point within the triangle. Calculate the expected value of X and Y, i.e., expected co-ordinates of a randomly chosen point. Find the correlation and covariance between X and Y. 10 (CO1) (PO1)

   c) Suppose you are fishing in a pond where there are 3 red fish, 4 green fish, and 5 blue fish. If you use a net to scoop up 6 of them, find the probability that you scooped up exactly 2 of each. 10 (CO1) (PO1)

   If you catch and release the fishes until you caught 6 fishes (catch 1, throw it back, catch another, throw it back, etc.), find the probability that you caught exactly 2 of each?

2. a) A mouse is trapped in a room with three exits at the centre of a maze. Exit 1 leads outside the maze after 3 minutes. Exit 2 leads back to the room after 5 minutes. Exit 3 leads back to the room after 7 minutes. Every time the mouse makes a choice, it is equally likely to choose any of the three exits (Fig. 1). Find the expected time required for the mouse to leave the maze for good. 10 (CO2) (PO2)

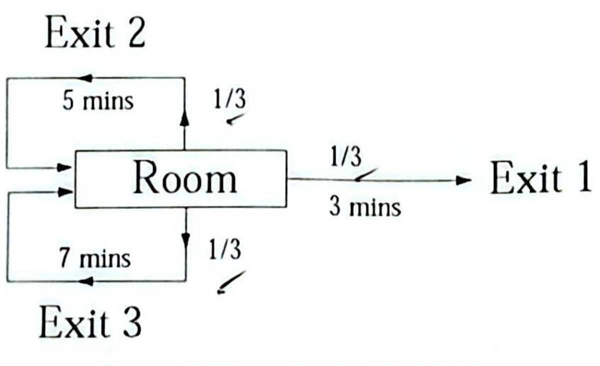


Figure 1: Figure for Question 2 (a)

   b) Suppose the mouse is put in a new maze with two rooms, as shown in Fig. 2. Starting from Room 1, find the expected number of attempts the mouse takes before it reaches the Exit (i.e., outside). Note that an attempt is a selection of choices for the mouse and leads to its next location. 10 (CO2) (PO2)

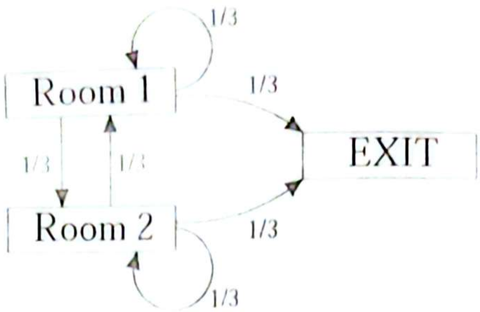


Figure 2: Figure for Question 2 (b)

c) With a modification to Question 2(b), assume that the mouse is initially put in Room 1 with probability $\frac{2}{3}$ and in Room 2 with probability $\frac{1}{3}$. Find the expected number of attempts the mouse takes before it reaches the Exit (i.e., outside). 10 (CO2) (PO2)

3. a) When sending messages over a network there is a chance that the bits will become corrupt. A Hamming Code allows for a 4-bit code to be encoded as 7-bit code (along with the original 4 bits additional 3 bits are sent), and maintains the property that if 0 or 1 bit is corrupted then the message can be perfectly reconstructed. Suppose that the probability of any bit being corrupted is 0.1. 10 (CO2) (PO2)

   Find the probabilities that a random message is received successfully with and without the Hamming code.

b) Alex decided he wanted to create a new type of distribution that will be famous. He knows he wants it to be continuous and have uniform density. He will denote a random variable X having the Uniform-2 distribution as $X \sim$ Uniform-2(a,b,c,d), where $a < b < c < d$. He wants the density to be non-zero in $[a,b]$ and $[c,d]$, and zero everywhere else. Anywhere the density is non-zero, it must be equal to the same constant. 10 (CO2) (PO2)

   Find the probability density function $f_X(x)$ and the cumulative distribution function $F_X(x)$ of X.

c) A rare disease is found in 0.5% of a population. A two-stage test is developed to diagnose the disease. Stage 1 gives a positive result with probability 0.92 if a person has the disease, while it gives a positive result for a healthy person with probability 0.12. In contrast, stage 2 gives a positive result with probability 0.97 if the person has the disease, whereas it gives a positive result for a healthy person with probability 0.05. A person is tested stage 2 only if the stage 1 result is positive. 10 (CO2) (PO2)

   Find the probability that a random person is tested positive twice. If a random person is tested positive twice, find the probability that he or she has the disease.

4. a) In response to the increasing weight of airline passengers, the Federal Aviation administration in 2003 told airlines to assume that passengers average weight is 190 pounds in the summer, including clothing and carry-on baggage. However, the weights of the passengers vary, and the FAA did not specify a standard deviation. A reasonable standard deviation is 35 pounds. Weight are not normally distributed (especially when the population includes both men and women), but they are close to Normal. A commuter plane carries 19 passengers. What is the approximate probability that the average weight of a passenger is between 170 and 210 pounds. Also, find the approximate probability that the total weight of the passengers exceeds 4000 pounds. 10 (CO3) (PO4)

b) Let $X_1, X_2, \ldots, X_N$ be the sample drawn independently from a uniform distribution over a diamond-shaped area with edge length $\sqrt{2}\theta$ in $\mathbb{R}^2$, where $\theta \in \mathbb{R}^+$ (see Figure 3). Therefore, $X_i \in \mathbb{R}^2$, and the distribution is 10 (CO3) (PO4)

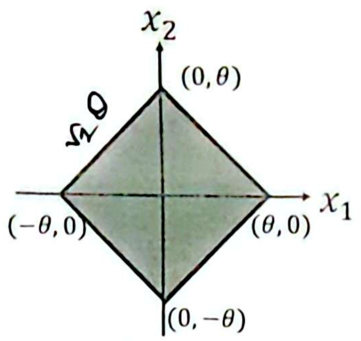


Figure 3: Figure for Question 4 (b)

$$f_X(x|\theta) = \begin{cases} \frac{1}{2\theta^2}, & \text{for } ||x|| \le \theta; \\ 0, & \text{otherwise.} \end{cases}$$

where $||x|| = |x_1| + |x_2|$ is the L_1 norm. Find the maximum likelihood estimate of θ.

5. a) A machine is set to fill pots with yogurt such that the mean weight of yogurt in a pot is 505 grams. To check that the machine is working properly, a random sample of 8 pots is selected. The weight of yogurt, in grams, in each pot is as follows

508 510 500 500 498 503 508 505

Suppose the weights of the yogurt delivered by the machine follow a normal distribution,

i. Assuming the standard deviation is 5.4 grams, find a 95% confidence interval for the mean weight, μ grams, of yogurt in a pot. 7 (CO3) (PO4)

ii. Assuming that the standard deviation is unknown, find a 98% confidence interval for the mean weight, μ grams, of yogurt in a pot. 7 (CO3) (PO4)

b) A statistician chooses 27 randomly selected dates, and when examining the occupancy records of a particular motel for those dates, finds a standard deviation of 5.86 rooms rented. If the number of rooms rented is normally distributed, find the 95% confidence interval for the population standard deviation of the number of rooms rented. 6 (CO3) (PO4)

6. The lifetime, X hours, of Everwhite camera batteries is normally distributed. The manufacturer claims that the mean lifetime of these batteries is 100 hours.

a) The members of a photography club suspect that the batteries do not last as long as is claimed by the manufacturer. In order to investigate their suspicion, the members test a random sample of five of these batteries and find the lifetimes, in hours, to be as follows: 10 (CO3) (PO4)

85 92 100 95 99

Test the members' suspicion at the 5% level of significance. Also, find the p-value of the test.

b) The manufacturer, believing that the mean lifetime of these batteries has not changed from 100 hours, decides to determine the lifetime, x hours, each of a random sample of 80 Everwhite camera batteries. The manufacturer obtains the following results, where $\bar{x}$ denotes the sample mean: 10 (CO3) (PO4)

$$\sum x = 8080 \qquad \text{and} \qquad \sum(x-\bar{x})^2 = 6399$$

Test the manufacturer's belief at the 5% level of significance. For a large sample size, you can assume that the sample approximately has the normal distribution.

Appendix - A

PMF/PDF, expected values and variance of known Random Variables

Families of Distribution	PMF or PDF	Expectation	Variance
Bernoulli $X \sim Ber(p)$	$P_X(x) = \begin{cases} p^x(1-p)^{1-x}, & x = 0,1 \\ 0, & \text{otherwise.} \end{cases}$	p	$p(1-p)$
Geometric $X \sim Geom(p)$	$P_X(x) = \begin{cases} p(1-p)^{x-1}, & x = 1,2,\ldots \\ 0, & \text{otherwise.} \end{cases}$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$
Binomial $X \sim Bin(n,p)$	$P_X(x) = \begin{cases} \binom{n}{x} p^x(1-p)^{n-x}, & x = 0,1,\ldots,n \\ 0, & \text{otherwise.} \end{cases}$	np	$np(1-p)$
Pascal/-ve Binomial $X \sim pascal(k,p)$	$P_X(x) = \begin{cases} \binom{x-1}{k-1} p^k(1-p)^{x-k}, & x = k, k+1,\ldots \\ 0, & \text{otherwise.} \end{cases}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$
Poisson $X \sim Poisson(\lambda)$	$P_X(x) = \begin{cases} \frac{(\lambda T)^x e^{-(\lambda T)}}{x!}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	λT	λT
Hyper Geometric $X \sim HGem(r,g,n)$	$P_X(x) = \begin{cases} \frac{\binom{r}{x}\binom{g}{n-x}}{\binom{r+g}{n}}, & x = 0,1,\ldots,\min(r,n) \\ 0, & \text{otherwise.} \end{cases}$	$\frac{rn}{r+g}$	$\frac{nrg}{(r+g)^2}(1-\frac{n-1}{r+g-1})$
Uniform (discrete) $X \sim unif(a,b)$	$P_X(x) = \begin{cases} \frac{1}{b-a+1}, & x = a, a+1,\ldots,b \\ 0, & \text{otherwise.} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)(b-a+2)}{12}$
Exponential $X \sim exp(\lambda)$	$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gaussian $X \sim N(\mu,\sigma^2)$	$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma}}, & -\infty < x < +\infty \\ 0, & \text{otherwise.} \end{cases}$	μ	σ^2
Uniform (Continuous) $X \sim unif(a,b)$	$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise.} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Gamma $X \sim gam(r,\lambda)$	$f_X(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{r-1}}{\Gamma(r)}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$
Multinomial	$P_{X_1,\ldots,X_r}(x_1,\ldots,x_r) = \begin{cases} \binom{n}{x_1 \ldots x_r} p_1^{x_1} \ldots p_r^{x_r}, & x_i = 0,\ldots,n, 0 \leq i \leq r, \sum_{i=0}^{r} x_i = n \\ 0, & \text{otherwise.} \end{cases}$		
Multivariate Hyper geometric	$P_{X_1,\ldots,X_r}(x_1,\ldots,x_r) = \begin{cases} \frac{\binom{n_1}{x_1}\binom{n_2}{x_2}\cdots\binom{n_r}{x_r}}{\binom{n_1+\cdots+n_r}{x_1+\cdots+x_r}}, & x_i = 0,1,\ldots,\min(r,n_i), r = x_1 + \cdots + x_r \\ 0, & \text{otherwise.} \end{cases}$		

Appendix - B

General Formulas

Name	Formula
Conditional Probability	$P[A\|B] = \frac{P[AB]}{P[B]}, P[B] > 0$
Product Rule	$P[AB] = P[A]P[B\|A]$ $P[A_1 \dots A_n] = P[A_1]P[A_2\|A_1]\dots P[A_n\|A_1 \dots A_{n-1}]$
Conditional Product Rule	$P[AB\|C] = P[A\|C]P[B\|AC]$
Sum Rule	$P[A] = \sum_{i=1}^{n} P[A\|B_i]P[B_i]$
Conditional Sum Rule	$P[A\|C] = \sum_{i=1}^{n} P[A\|B_i, C]P[B_i\|C]$
Bayes' Theorem	$P[B_i\|A] = \frac{P[A\|B_i]P[B_i]}{\sum_{i=1}^{n} P[A\|B_i]P[B_i]}$
Conditional Bayes' Theorem	$P[B_i\|A, C] = \frac{P[A\|B_i, C]P[B_i\|C]}{\sum_{i=1}^{n} P[A\|B_i, C]P[B_i\|C]}$
Independence of Events	$P[AB] = P[A]P[B]$
Expected Value	$E[X] = \sum_{x \in S_X} xP_X(x)$ $E[X] = \int_{-\infty}^{+\infty} x f_X(x)dx$
Variance	$Var[X] = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$
Marginal PMF	$P_X(x) = \sum_{x \in S_X} P_{XY}(x, y)$ $f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y)dy$
Covariance	$Cov[X, Y] = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$
Correlation	$r_{XY} = E[XY]$
Correlation Coefficient	$\rho_{XY} = \frac{Cov[X,Y]}{\sqrt{Var[X]Var[Y]}}$
Conditional Distribution	$P_{X\|Y}(x\|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$, $f_{Y\|X}(y\|x) = \frac{f_{XY}(x,y)}{f_X(x)}$
Product Rule	$P_{XY}(x, y) = P_{X\|Y}(x\|y)P_Y(y)$, $f_{XY}(x, y) = f_{Y\|X}(y\|x)f_X(x)$
Marginal PMF/PDF	$P_X(x) = \sum_y P_{XY}(x, y)$, $f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y)dy$
Sum Rule	$P_X(x) = \sum_y P_{X\|Y}(x\|y)P_Y(y)$, $f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y)f_Y(y)dy$
Conditional Expectation	$E[X\|Y = y] = \sum_x xP_{X\|Y}(x\|y)$, $E[X\|Y = y] = \int_{-\infty}^{+\infty} x f_{X\|Y}(x\|y)dx$
Law of Total Expectation	$E[X] = \sum_y E[X\|Y = y]P_Y(y)$, $E[Y] = \int_{-\infty}^{+\infty} E[Y\|X = x]f_X(x)dx$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \overline{X} + z_\alpha\frac{\sigma}{\sqrt{n}}\right)$	$\left(\overline{X} + z_\alpha\frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\overline{X} \pm t_{\alpha/2, n-1}\frac{S}{\sqrt{n}}$	$\left(-\infty, \overline{X} + t_{\alpha, n-1}\frac{S}{\sqrt{n}}\right)$	$\left(\overline{X} - t_{\alpha, n-1}\frac{S}{\sqrt{n}}, \infty\right)$
μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}\right)$	$\left(0, \frac{(n-1)S^2}{\chi^2_{1-\alpha, n-1}}\right)$	$\left(\frac{(n-1)S^2}{\chi^2_{\alpha, n-1}}, \infty\right)$

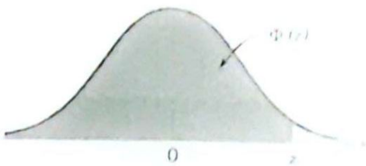
Table: 100(1-α) percent confidence interval

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $\|TS\| > z_{\alpha/2}$	$2P\{Z \geq \|t\|\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS > z_\alpha$	$P\{Z \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS < -z_\alpha$	$P\{Z \leq t\}$

Table: Test statistics and p-values for Z Distributed sample mean

Appendix-C: Standard Normal Distribution

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.500000	0.503989	0.507978	0.511967	0.515953	0.519939	0.532922	0.527903	0.531881	0.535856
0.1	0.539828	0.543795	0.547758	0.551717	0.555760	0.559618	0.563559	0.567495	0.571424	0.575345
0.2	0.579260	0.583166	0.587064	0.590954	0.594835	0.598706	0.602568	0.606420	0.610261	0.614092
0.3	0.617911	0.621719	0.625516	0.629300	0.633072	0.636831	0.640576	0.644309	0.648027	0.651732
0.4	0.655422	0.659097	0.662757	0.666402	0.670031	0.673645	0.677242	0.680822	0.684386	0.687933
0.5	0.691462	0.694974	0.698468	0.701944	0.705401	0.708840	0.712260	0.715661	0.719043	0.722405
0.6	0.725747	0.729069	0.732371	0.735653	0.738914	0.742154	0.745373	0.748571	0.751748	0.754903
0.7	0.758036	0.761148	0.764238	0.767305	0.770350	0.773373	0.776373	0.779350	0.782305	0.785236
0.8	0.788145	0.791030	0.793892	0.796731	0.799546	0.802338	0.805106	0.807850	0.810570	0.813267
0.9	0.815940	0.818589	0.821214	0.823815	0.826391	0.828944	0.831472	0.833977	0.836457	0.838913
1.0	0.841345	0.843752	0.846136	0.848495	0.850830	0.853141	0.855428	0.857690	0.859929	0.862143
1.1	0.864334	0.866500	0.868643	0.870762	0.872857	0.874928	0.876976	0.878999	0.881000	0.882977
1.2	0.884930	0.886860	0.888767	0.890651	0.892512	0.894350	0.896165	0.897958	0.899727	0.901475
1.3	0.903199	0.904902	0.906582	0.908241	0.909877	0.911492	0.913085	0.914657	0.916207	0.917736
1.4	0.919243	0.920730	0.922196	0.923641	0.925066	0.926471	0.927855	0.929219	0.930563	0.931888
1.5	0.933193	0.934478	0.935744	0.936992	0.938220	0.939429	0.940620	0.941792	0.942947	0.944083
1.6	0.945201	0.946301	0.947384	0.948449	0.949497	0.950529	0.951543	0.952540	0.953521	0.954486
1.7	0.955435	0.956367	0.957284	0.958185	0.959071	0.959941	0.960796	0.961636	0.962462	0.963273
1.8	0.964070	0.964852	0.965621	0.966375	0.967116	0.967843	0.968557	0.969258	0.969946	0.970621
1.9	0.971283	0.971933	0.972571	0.973197	0.973810	0.974412	0.975002	0.975581	0.976148	0.976705
2.0	0.977250	0.977784	0.978308	0.978822	0.979325	0.979818	0.980301	0.980774	0.981237	0.981691
2.1	0.982136	0.982571	0.982997	0.983414	0.983823	0.984222	0.984614	0.984997	0.985371	0.985738
2.2	0.986097	0.986447	0.986791	0.987126	0.987455	0.987776	0.988089	0.988396	0.988696	0.988989
2.3	0.989276	0.989556	0.989830	0.990097	0.990358	0.990613	0.990863	0.991106	0.991344	0.991576
2.4	0.991802	0.992024	0.992240	0.992451	0.992656	0.992857	0.993053	0.993244	0.993431	0.993613
2.5	0.993790	0.993963	0.994132	0.994297	0.994457	0.994614	0.994766	0.994915	0.995060	0.995201
2.6	0.995339	0.995473	0.995604	0.995731	0.995855	0.995975	0.996093	0.996207	0.996319	0.996427
2.7	0.996533	0.996636	0.996736	0.996833	0.996928	0.997020	0.997110	0.997197	0.997282	0.997365
2.8	0.997445	0.997523	0.997599	0.997673	0.997744	0.997814	0.997882	0.997948	0.998012	0.998074
2.9	0.998134	0.998193	0.998250	0.998305	0.998359	0.998411	0.998462	0.998511	0.998559	0.998605
3.0	0.998650	0.998694	0.998736	0.998777	0.998817	0.998856	0.998893	0.998930	0.998965	0.998999
3.1	0.999032	0.999065	0.999096	0.999126	0.999155	0.999184	0.999211	0.999238	0.999264	0.999289
3.2	0.999313	0.999336	0.999359	0.999381	0.999402	0.999423	0.999443	0.999462	0.999481	0.999499
3.3	0.999517	0.999533	0.999550	0.999566	0.999581	0.999596	0.999610	0.999624	0.999638	0.999650
3.4	0.999663	0.999675	0.999687	0.999698	0.999709	0.999720	0.999730	0.999740	0.999749	0.999758
3.5	0.999767	0.999776	0.999784	0.999792	0.999800	0.999807	0.999815	0.999821	0.999828	0.999835
3.6	0.999841	0.999847	0.999853	0.999858	0.999864	0.999869	0.999874	0.999879	0.999883	0.999888
3.7	0.999892	0.999896	0.999900	0.999904	0.999908	0.999912	0.999915	0.999918	0.999922	0.999925
3.8	0.999928	0.999931	0.999933	0.999936	0.999938	0.999941	0.999943	0.999946	0.999948	0.999950
3.9	0.999952	0.999954	0.999956	0.999958	0.999959	0.999961	0.999963	0.999964	0.999966	0.999967

Appendix-D: t Distribution

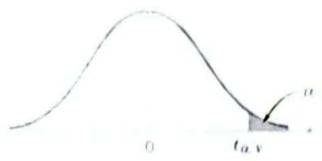


Table IV Percentage Points $t_{\alpha,\nu}$ of the t-Distribution

ν \ α	.40	.25	.10	.05	.025	.01	.005	.0025	.001	.0005
1	.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	.289	.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	.277	.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	.271	.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	.267	.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	.265	.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	.263	.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	.262	.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	.261	.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	.260	.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	.260	.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	.259	.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	.259	.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	.258	.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	.258	.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	.258	.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	.257	.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	.257	.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	.257	.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	.257	.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	.257	.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	.256	.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	.256	.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	.256	.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	.256	.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	.256	.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	.256	.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	.256	.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	.256	.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	.256	.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	.255	.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	.254	.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	.254	.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	.253	.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

ν = degrees of freedom.

Appendix E. Percentage points of the chi-Square distribution

Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81	140.17

ν = degrees of freedom.

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $\lvert TS\rvert > t_{\alpha/2,\, n-1}$	$2P\{T_{n-1} \geq \lvert t\rvert\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS > t_{\alpha,\, n-1}$	$P\{T_{n-1} \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS < -t_{\alpha,\, n-1}$	$P\{T_{n-1} \leq t\}$

Table: Test statistics and p-values for t_{n-1} Distributed sample mean